

1 Methods

This section describes the equations integrated by MuJoCo-Orbit, the frames in which they live, and how the chief-orbit propagator and the MuJoCo multibody/contact solver are coupled in a single step. We begin with the standard inertial equations of orbit mechanics, then derive the chief-centered equations that MuJoCo actually integrates, and finally specify the force model and coupling architecture.

1.1 Notation

We use the following symbols throughout. SI units are used inside MuJoCo; the orbital layer uses km, km/s, km/s².

$\mathbf{R} \in \mathbb{R}^3$ Chief reference position in the Earth-centered inertial (ECI) frame, in km.

$\mathbf{V} \in \mathbb{R}^3$ Chief reference velocity in ECI, in km/s.

$t \in \mathbb{R}$ Mission elapsed time, in s.

$\mathbf{r}_i \in \mathbb{R}^3$ Position of body i 's center of mass in the MuJoCo world frame (chief-centered, ECI-parallel axes), in m.

$\mathbf{v}_i \in \mathbb{R}^3$ Linear velocity of body i 's COM in the world frame, in m/s.

$\boldsymbol{\omega}_i \in \mathbb{R}^3$ Angular velocity of body i , expressed in the body frame, in rad/s.

$\mathbf{R}_i = \mathbf{R} + 10^{-3}\mathbf{r}_i$ Absolute ECI position of body i , in km.

m_i Mass of body i , in kg.

$J_i \in \mathbb{R}^{3 \times 3}$ Inertia tensor of body i about its COM, in body frame, in kg m².

$\mu = GM_\oplus$ Earth standard gravitational parameter, 398 600.4418 km³/s².

$R_\oplus$ Earth equatorial radius, 6378.137 km.

J_2 Earth second zonal coefficient, 1.082626×10^{-3} .

$\mathbf{B}(\mathbf{R}, t)$ Geomagnetic flux density at $\mathbf{R}$, in T.

$\rho(\mathbf{R})$ Atmospheric mass density at $\mathbf{R}$, in kg/m³.

$\boldsymbol{\omega}_\oplus$ Earth angular velocity vector in ECI, $\boldsymbol{\omega}_\oplus = (0, 0, \omega_\oplus)$.

C_{IL}, C_{LI} Direction cosine matrices ECI $\leftrightarrow$ LVLH; C_{LI} maps ECI vectors to LVLH coordinates and $C_{IL} = C_{LI}^\top$.

R_i^{wb} Rotation matrix world-from-body for body i (MuJoCo `xmat`); R_i^{wp} is world-from-principal-axes (MuJoCo `ximat`).

We denote the absolute ECI velocity of body i by $\mathbf{V}_i = \mathbf{V} + 10^{-3}\mathbf{v}_i$.

1.2 Frames

Three frames are used.

ECI ($\mathcal{F}_I$) A non-rotating geocentric inertial frame. The chief reference state $(\mathbf{R}, \mathbf{V}, t)$ and all environmental fields $(\mathbf{B}, \rho, \text{sun direction } \hat{\mathbf{s}})$ are expressed here.

World ($\mathcal{F}_W$) A chief-centered, ECI-axis-parallel frame. Its origin follows $\mathbf{R}(t)$, its axes are parallel to ECI axes, and consequently its angular velocity in $\mathcal{F}_I$ is zero. This is the frame in which MuJoCo represents $\mathbf{r}_i$, $\mathbf{v}_i$, R_i^{wb} , and all applied wrenches. $\mathcal{F}_W$ is non-inertial because its origin accelerates, but it is non-rotating: only a translational pseudo-force appears in it (Section 1.5).

LVLH ($\mathcal{F}_L$) A chief-centered rotating frame defined by

$$\hat{\mathbf{x}}_L = \frac{\mathbf{R}}{\|\mathbf{R}\|}, \quad \hat{\mathbf{z}}_L = \frac{\mathbf{R} \times \mathbf{V}}{\|\mathbf{R} \times \mathbf{V}\|}, \quad \hat{\mathbf{y}}_L = \hat{\mathbf{z}}_L \times \hat{\mathbf{x}}_L, \quad (1)$$

with angular velocity (in LVLH coordinates)

$$\boldsymbol{\omega}_L = C_{LI} \frac{\mathbf{R} \times \mathbf{V}}{\|\mathbf{R}\|^2}. \quad (2)$$

$\mathcal{F}_L$ is a derived frame used for analysis and IO; it is not the frame in which MuJoCo integrates.

1.3 Orbit Mechanics: Inertial Equations

For a point mass at ECI position $\mathbf{R}$, the Newtonian equation of motion under Earth's gravity field plus arbitrary perturbations is

$$\ddot{\mathbf{R}} = \mathbf{g}(\mathbf{R}) + \mathbf{a}^{\text{pert}}(\mathbf{R}, \mathbf{V}, t), \quad (3)$$

where $\mathbf{g}(\mathbf{R})$ is the gravitational acceleration and $\mathbf{a}^{\text{pert}}$ aggregates non-gravitational perturbations.

Two-body gravity

$$\mathbf{g}_{2B}(\mathbf{R}) = -\frac{\mu}{\|\mathbf{R}\|^3} \mathbf{R}. \quad (4)$$

J_2 **correction** The dominant zonal harmonic adds

$$\mathbf{g}_{J_2}(\mathbf{R}) = -\frac{3}{2} \mu J_2 \frac{R_\oplus^2}{\|\mathbf{R}\|^5} \begin{bmatrix} (1 - 5(R_z/\|\mathbf{R}\|)^2) R_x \\ (1 - 5(R_z/\|\mathbf{R}\|)^2) R_y \\ (3 - 5(R_z/\|\mathbf{R}\|)^2) R_z \end{bmatrix}, \quad (5)$$

so the gravity model used by MuJoCo-Orbit is $\mathbf{g}(\mathbf{R}) = \mathbf{g}_{2B}(\mathbf{R}) + \mathbf{g}_{J_2}(\mathbf{R})$ when J_2 is enabled, and $\mathbf{g}_{2B}(\mathbf{R})$ otherwise.

1.4 Chief Orbit Propagation

The chief is integrated as a point mass under gravity plus a swarm-averaged non-gravity feedback acceleration $\mathbf{a}_{\text{fb}}$ (defined in Section 1.5.2):

$$\frac{d}{dt} \begin{bmatrix} \mathbf{R} \\ \mathbf{V} \end{bmatrix} = \begin{bmatrix} \mathbf{V} \\ \mathbf{g}(\mathbf{R}) + \mathbf{a}_{\text{fb}} \end{bmatrix}. \quad (6)$$

Equation (6) is integrated by classical fourth-order Runge–Kutta with step size Δt_o (default: equal to the MuJoCo timestep Δt),

$$\mathbf{k}_1 = \mathbf{f}(\mathbf{R}, \mathbf{V}), \quad (7)$$

$$\mathbf{k}_2 = \mathbf{f}\left(\mathbf{R} + \frac{\Delta t_o}{2} \mathbf{k}_1^R, \mathbf{V} + \frac{\Delta t_o}{2} \mathbf{k}_1^V\right), \quad (8)$$

$$\mathbf{k}_3 = \mathbf{f}\left(\mathbf{R} + \frac{\Delta t_o}{2} \mathbf{k}_2^R, \mathbf{V} + \frac{\Delta t_o}{2} \mathbf{k}_2^V\right), \quad (9)$$

$$\mathbf{k}_4 = \mathbf{f}\left(\mathbf{R} + \Delta t_o \mathbf{k}_3^R, \mathbf{V} + \Delta t_o \mathbf{k}_3^V\right), \quad (10)$$

$$\begin{bmatrix} \mathbf{R} \\ \mathbf{V} \end{bmatrix}_{n+1} = \begin{bmatrix} \mathbf{R} \\ \mathbf{V} \end{bmatrix}_n + \frac{\Delta t_o}{6} (\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4). \quad (11)$$

Here $\mathbf{f}(\mathbf{R}, \mathbf{V}) = (\mathbf{V}, \mathbf{g}(\mathbf{R}) + \mathbf{a}_{\text{fb}})$ packs the right-hand side of (6). The chief acceleration appearing later in the chief-frame equations is therefore

$$\mathbf{a}_{\text{chief}} \equiv \ddot{\mathbf{R}} = \mathbf{g}(\mathbf{R}) + \mathbf{a}_{\text{fb}}. \quad (12)$$

1.5 Multi-Body Equations of Motion in the Chief Frame

Let body i have absolute position $\mathbf{R}_i = \mathbf{R} + 10^{-3} \mathbf{r}_i$. Newton in $\mathcal{F}_I$ reads

$$m_i \ddot{\mathbf{R}}_i = m_i \mathbf{g}(\mathbf{R}_i) + \mathbf{F}_i^{\text{ng}}, \quad (13)$$

with $\mathbf{F}_i^{\text{ng}}$ the non-gravity force on body i . Substituting $\mathbf{R}_i = \mathbf{R} + 10^{-3} \mathbf{r}_i$ and rearranging,

$$m_i 10^{-3} \ddot{\mathbf{r}}_i = m_i [\mathbf{g}(\mathbf{R}_i) - \mathbf{a}_{\text{chief}}] + \mathbf{F}_i^{\text{ng}}. \quad (14)$$

Converting the $\mathbf{g}$ -difference from km/s^2 to m/s^2 ($\times 10^3$) and dividing both sides by 10^{-3} ,

$$m_i \ddot{\mathbf{r}}_i = \underbrace{m_i [\tilde{\mathbf{g}}(\mathbf{R}_i) - \tilde{\mathbf{g}}(\mathbf{R})]}_{\text{differential gravity}} - \underbrace{m_i \tilde{\mathbf{a}}_{\text{fb}}}_{\text{origin compensation}} + \mathbf{F}_i^{\text{ng}}, \quad (15)$$

where tildes denote SI quantities ($\tilde{\mathbf{g}} \equiv 10^3 \mathbf{g}$, $\tilde{\mathbf{a}}_{\text{fb}} \equiv 10^3 \mathbf{a}_{\text{fb}}$). Equation (15) is the equation of motion that MuJoCo integrates per body. Because $\mathcal{F}_W$ is non-rotating, no Coriolis, centrifugal, or Euler terms appear; the only fictitious force is the translational $-m_i \mathbf{a}_{\text{chief}}$, which is split into the gravity piece $-m_i \mathbf{g}(\mathbf{R})$ (absorbed into the differential gravity term) and the non-gravity piece $-m_i \mathbf{a}_{\text{fb}}$ (the origin compensation).

1.5.1 Per-body external force model

The non-gravity force $\mathbf{F}_i^{\text{ng}}$ is the sum of all environmental and actuator contributions, evaluated at body i 's own state:

$$\mathbf{F}_i^{\text{ng}} = \mathbf{F}_i^{\text{drag}} + \mathbf{F}_i^{\text{SRP}} + \mathbf{F}_i^{\text{thr}}. \quad (16)$$

Magnetic, magnetorquer, RW, CMG, and gravity-gradient effects produce torques only and are summarised below.

Drag For each surface k on body i with body-frame normal $\hat{\mathbf{n}}_k$, area A_k , drag coefficient C_d^k , and center of pressure $\mathbf{p}_k$, denote the surface point ECI position and velocity by $\mathbf{R}_{ik} = \mathbf{R} + 10^{-3}(\mathbf{r}_i + R_i^{wb} \mathbf{p}_k)$ and $\mathbf{V}_{ik} = \mathbf{V} + 10^{-3}(\mathbf{v}_i + R_i^{wb}(\boldsymbol{\omega}_i \times \mathbf{p}_k))$. The atmosphere co-rotates with Earth, so the relative velocity at the surface is

$$\mathbf{V}_{ik}^{\text{rel}} = \mathbf{V}_{ik} - \boldsymbol{\omega}_{\oplus} \times \mathbf{R}_{ik}, \quad (17)$$

and the drag force in world coordinates is

$$\mathbf{F}_{ik}^{\text{drag}} = -\frac{1}{2} \rho(\mathbf{R}) C_d^k A_k \max(0, \hat{\mathbf{n}}_k^w \cdot \hat{\mathbf{V}}^{\text{rel}}) \|\mathbf{V}_{ik}^{\text{rel}}\| \mathbf{V}_{ik}^{\text{rel}}, \quad (18)$$

with $\hat{\mathbf{n}}_k^w = R_i^{wb} \hat{\mathbf{n}}_k$, $\hat{\mathbf{V}}^{\text{rel}} = \mathbf{V}_{ik}^{\text{rel}} / \|\mathbf{V}_{ik}^{\text{rel}}\|$ (positive only on the windward face). The accompanying torque about body i 's COM is

$$\boldsymbol{\tau}_{ik}^{\text{drag}} = (R_i^{wb} \mathbf{p}_k) \times \mathbf{F}_{ik}^{\text{drag}}. \quad (19)$$

$\rho(\mathbf{R})$ is currently evaluated at the chief position only and shared across surfaces; per-body velocity is the dominant term for short-baseline swarms.

Solar radiation pressure For an unshadowed surface ($\eta(\mathbf{R}) > 0$, with eclipse factor $\eta \in \{0, 1\}$),

$$\mathbf{F}_{ik}^{\text{SRP}} = -\eta P_{\odot} C_r^k A_k \max(0, \hat{\mathbf{n}}_k^w \cdot \hat{\mathbf{s}}) \hat{\mathbf{s}}, \quad (20)$$

where $P_{\odot} = 4.56 \times 10^{-6} \text{ N/m}^2$ is the solar pressure at 1 AU and $\hat{\mathbf{s}}$ is the Sun unit vector in ECI.

Residual magnetic dipole A body-frame residual dipole $\mathbf{m}_i^b$ produces

$$\boldsymbol{\tau}_i^{\text{mag}} = R_i^{wb} (\mathbf{m}_i^b \times R_i^{bw} \mathbf{B}(\mathbf{R})), \quad (21)$$

with $R_i^{bw} = (R_i^{wb})^{\top}$. $\mathbf{B}$ is a tilted dipole evaluated at the chief position.

Magnetorquers Each magnetorquer commands a body-frame dipole $\mathbf{m}_i^{b,\text{cmd}}$ subject to a magnitude limit m_{max} , producing $\boldsymbol{\tau} = \mathbf{m}^{b,\text{cmd}} \times \mathbf{B}^b$ in the body frame and rotated to world.

Gravity-gradient torque The differential-gravity force in (15) is evaluated at body i 's COM and therefore produces no net torque on a single rigid body. The analytic torque from integrating gravity over the body's mass distribution, in the small-extent limit $\|\mathbf{r}_i\| \ll \|\mathbf{R}_i\|$, is

$$\boldsymbol{\tau}_i^{gg} = \frac{3\mu}{\|\mathbf{R}_i\|^3} \hat{\mathbf{R}}_i \times (J_i^w \hat{\mathbf{R}}_i), \quad \hat{\mathbf{R}}_i = \mathbf{R}_i / \|\mathbf{R}_i\|, \quad (22)$$

where $J_i^w = R_i^{wp} \text{diag}(I_{i,1}, I_{i,2}, I_{i,3}) (R_i^{wp})^{\top}$ is the inertia tensor in world axes built from the principal moments and the world-from-principal-axes rotation R_i^{wp} (MuJoCo `ximat`).

Reaction wheels For wheel j with axis $\hat{\mathbf{a}}_j^b$, inertia I_j , and signed wheel speed Ω_j , the wheel angular momentum in the body frame is $\mathbf{h}_j^b = I_j \Omega_j \hat{\mathbf{a}}_j^b$. The torque on the host body is the gyroscopic reaction plus a commanded acceleration α_j :

$$\boldsymbol{\tau}_j^{rw,b} = -\boldsymbol{\omega}_i \times \mathbf{h}_j^b - I_j \alpha_j \hat{\mathbf{a}}_j^b, \quad (23)$$

with $\alpha_j = \tau_j^{\text{cmd}} / I_j$ saturated by torque/speed limits. The wheel speed is integrated separately by Euler: $\Omega_j^{n+1} = \Omega_j^n + \alpha_j \Delta t$.

Single-gimbal CMG With gimbal axis $\hat{\mathbf{g}}^b$, initial spin axis $\hat{\mathbf{s}}^b(0)$, rotor momentum h , and gimbal angle θ_j ,

$$\hat{\mathbf{s}}^b(\theta) = \cos \theta \hat{\mathbf{s}}^b(0) + \sin \theta \hat{\mathbf{g}}^b \times \hat{\mathbf{s}}^b(0), \quad (24)$$

$$\boldsymbol{\tau}_j^{\text{cmg},b} = -\boldsymbol{\omega}_i \times (h \hat{\mathbf{s}}^b(\theta)) - h \dot{\theta}_j \hat{\mathbf{g}}^b \times \hat{\mathbf{s}}^b(\theta), \quad (25)$$

with gimbal angle integrated by Euler under rate and angle limits.

Thrusters A thruster on body i at body-frame position $\mathbf{p}^b$ along direction $\hat{\mathbf{d}}^b$ with commanded force $F_j \in [0, F_{\max}]$ produces world-frame force and torque

$$\mathbf{F}_j^w = R_i^{wb} F_j \hat{\mathbf{d}}^b, \quad \boldsymbol{\tau}_j^w = R_i^{wb} ((\mathbf{p}^b - \mathbf{p}^{b,com}) \times F_j \hat{\mathbf{d}}^b). \quad (26)$$

1.5.2 Swarm-averaged feedback

The non-gravity translational force on body i contributes to a single shared feedback acceleration that drives the chief reference. Writing $M = \sum_i m_i$, define

$$\tilde{\mathbf{a}}_{\text{fb}} = \frac{1}{M} \sum_i \mathbf{F}_i^{\text{ng,trans}}, \quad (27)$$

where $\mathbf{F}_i^{\text{ng,trans}}$ is the translational portion of $\mathbf{F}_i^{\text{ng}}$ (drag, SRP, thrust). Converting to km/s^2 gives the $\mathbf{a}_{\text{fb}}$ used in (6). The same $\tilde{\mathbf{a}}_{\text{fb}}$ is subtracted from each body in (15) as the origin compensation $-m_i \tilde{\mathbf{a}}_{\text{fb}}$. This couples the chief reference to the swarm centroid without altering relative motion: any choice of chief trajectory yields the same $\mathbf{r}_i(t)$, because (15) is invariant under $\mathbf{a}_{\text{chief}} \rightarrow \mathbf{a}_{\text{chief}} + \mathbf{a}^*$ when the same $\mathbf{a}^*$ is applied to all bodies. The swarm-mean choice keeps $\|\mathbf{r}_i\|$ small.

1.6 MuJoCo Multibody and Contact Solver

Inside $\mathcal{F}_W$, MuJoCo treats $(\mathbf{r}_i, \dot{\mathbf{r}}_i, \mathbf{q}^{joint}, \dot{\mathbf{q}}^{joint})$ as a generalised state and integrates the standard articulated-body equations

$$M(q)\ddot{q} + \mathbf{c}(q, \dot{q}) = \boldsymbol{\tau}^{\text{passive}} + \boldsymbol{\tau}^{\text{ctrl}} + J_c^\top \boldsymbol{\lambda}, \quad (28)$$

where M is the joint-space mass matrix, $\mathbf{c}$ collects Coriolis, centripetal, and gyroscopic terms internal to the multibody chain (i.e. those generated by relative motion of links, not by frame rotation), $\boldsymbol{\tau}^{\text{passive}}$ is the passive force vector (`qfrc_passive`), $\boldsymbol{\tau}^{\text{ctrl}}$ is from MuJoCo actuators, and $J_c^\top \boldsymbol{\lambda}$ is the contact constraint impulse computed by MuJoCo's convex contact solver. MuJoCo's built-in uniform gravity is disabled (`opt.gravity = 0`) so that the gravity model in (15) is not duplicated.

MuJoCo-Orbit writes the per-body wrench $(\mathbf{F}_i^{\text{tot}}, \boldsymbol{\tau}_i^{\text{tot}})$ from Section 1.5.1 into $\boldsymbol{\tau}^{\text{passive}}$ via `mj_applyFT` at each body's COM. MuJoCo is otherwise responsible for: integrating $(q, \dot{q})$, enforcing joint and equality constraints, detecting contacts, and computing contact impulses $\boldsymbol{\lambda}$.

1.7 Coupling Architecture

The chief propagator and MuJoCo are coupled through a MuJoCo plugin (`mujoco_orbit.orbit`) that owns the chief state, environment caches, actuator state, and per-body wrench buffer. The plugin registers two capability hooks: `mjPLUGIN_PASSIVE` and a per-step `Advance` callback. A single call to `mjo_step` executes the following stages.

1. **Cache refresh.** Recompute LVLH bases $C_{IL}, C_{LI}, \boldsymbol{\omega}_L, \dot{\boldsymbol{\omega}}_L$ from $(\mathbf{R}, \mathbf{V})$ using (1)–(2) (with $\dot{\boldsymbol{\omega}}_L$ derived from $\mathbf{g}(\mathbf{R})$ including J_2 when enabled). Recompute $\hat{\mathbf{s}}(t)$, $\boldsymbol{\eta}(\mathbf{R}, t)$, $\mathbf{B}(\mathbf{R}, t)$, $\boldsymbol{\rho}(\mathbf{R})$.
2. **Force assembly** (Compute, passive stage). Evaluate (15)–(26) per body, accumulating into `qfrc_passive` via `mj_applyFT` and into a Python visible buffer.
3. **Multibody integration.** MuJoCo solves (28) and advances $(q, \dot{q})$ by Δt using its semi-implicit Euler or RK4 integrator (configurable in the XML).

4. **Actuator advance** (**Advance**). Euler-integrate reaction-wheel speeds Ω_j and CMG gimbal angles θ_j under saturation limits.
5. **Chief advance** (**Advance**). Compute $\mathbf{a}_{\text{fb}}$ from accumulated swarm forces (27), then integrate $(\mathbf{R}, \mathbf{V})$ over Δt_o using RK4 (11) on (6).
6. **Cache refresh**. Re-sync caches at the new $(\mathbf{R}, \mathbf{V})$ so subsequent reads of `data.frame` and `data.env` are consistent with the post-step chief state.

MuJoCo’s contact solver receives no special treatment: contacts are resolved in $\mathcal{F}_W$, where the only forces are the per-body wrenches above and MuJoCo’s internal joint and constraint forces. Contact impulses $\boldsymbol{\lambda}$ do not feed back into the chief orbit ($\mathbf{a}_{\text{fb}}$ uses external translational forces only, not contact impulses), preserving the interpretation of the chief as a free-flying reference.